# PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 5: Introduction to Practice Project 2 (Growth Curves, Chp. 4 of [Textbook])

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## Scientific context

### Agriculture and the birth of modern statistics

Agricultural and biological research played a major role in the development of modern statistics.

- Researchers needed to compare crop varieties, fertilizers, and growing conditions.
- Crop yields varied because of soil, weather, and other environmental factors.
- Reliable conclusions required distinguishing systematic effects from natural variation.
- These challenges stimulated the development of statistical inference and experimental design.

Many statistical methods used today originated in attempts to solve practical biological and agricultural problems.

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Ronald A. Fisher (1890–1962) worked at Rothamsted Experimental Station, one of the oldest agricultural research institutions in the world, in England from 1919 to 1933. His agricultural research contributed to the development of:

- Analysis of variance (ANOVA)
- Factorial experimental designs
- Modern methods of statistical inference (e.g., hypothesis testing)

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*Arabidopsis thaliana*, commonly called thale cress, is a small flowering plant in the mustard family (Brassicaceae). It is one of the most widely used model organisms in plant biology, particularly in genetics, development, and responses to the environment.



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Plant growth is a individualized dynamic biological process. A single height measurement cannot fully describe how a plant develops.

- Seeds emerge at different times after planting.
- Growth eventually slows and approaches a plateau.
- Plant height changes nonlinearly over time.
- Plants reach the mature heights at different times.

Repeated measurements provide information about these aspects of development.

## Scientific question

The scientific objective is to investigate whether genetic differences between the two groups ("strains") are associated with differences in their growth trajectories.

### What does a difference in growth mean?

Possible questions include:

1. Does strain "108" emerge earlier?
2. Does strain "108" reach a greater mature height?
3. Do strain "108" grows faster after emergence?

These questions are related, but not equivalent.

**Which comparison best addresses the original scientific objective?**

## Study design

### Repeated measurements of plant height

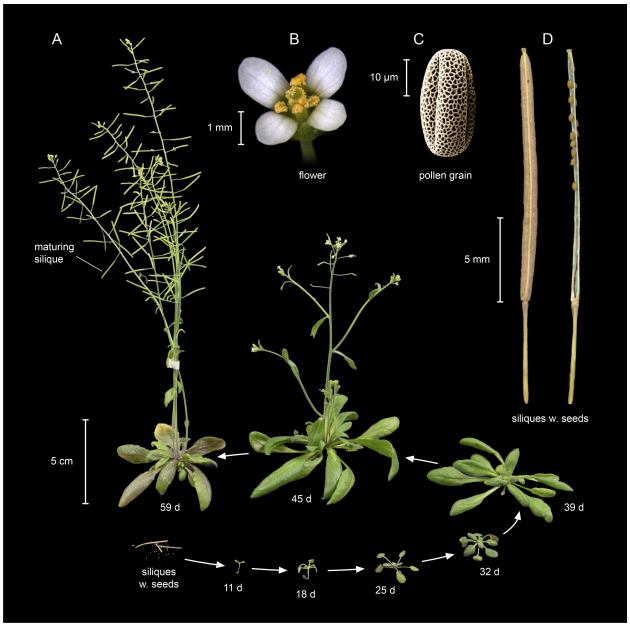
- Total: 70 plants
  - Strain `cis`: 40 plants
  - Strain `108`: 30 plants
- All seeds planted on the same date



- Heights recorded every four days
  - Plant height refers to the height of the visible plant above the soil surface, measured in millimeters.

Each plant contributes a sequence of measurements.

**This is a longitudinal study.**




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**Measurement schedule**

Days since planting:

| | | | | | |
|---|---|---|---|---|---|
| 29 | 33 | 37 | 41 | 45 | 49 |
| 53 | 57 | 61 | 65 | 69 | |

Eleven measurements per plant.

- No visible plants at day 29

- Plants emerge at different times
- All plants are visible by day 41
- Recorded heights reach a plateau by day 65

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**Two definitions of time**

**Calendar time**

Number of days since planting.

- Common time origin for all plants
- Includes differences in emergence timing

**Plant age**

Number of days since estimated emergence.

- Aligns plants by developmental stage
- Focuses on growth after emergence

**The choice of time origin changes the scientific comparison.**

**Emergence is not observed exactly**

Suppose a plant has:

| Day | Height (mm) |
|---|---|
| 33 | 0 |
| 37 | 3 |
| 41 | 8 |

Then, the plant emerged sometime between days 33 and 37.

The textbook uses the last observed zero-height day as the estimated emergence day.

**This is a convention, not an exact observation.**

## Final data

**PlantGrowth.dat**

https://www.stat.uchicago.edu/~pmcc/projects/ch04/PlantGrowth.dat

The file contains:

- Plant identifier within strain
- Strain indicator (“0” for “cis” and “1” for “108”)
- Eleven height measurements per plant

**Data organization**

**Wide format**

| Plant | Strain | Day 29 | Day 33 | ... | Day 69 |
|---|---|---|---|---|---|
| 1 | cis | 0 | 0 | ... | 24 |
| 2 | cis | 0 | 0 | ... | 23 |
| ... | ... | ... | ... | ... | ... |

**Long format**

| Plant | Strain | Day | Height |
|---|---|---|---|
| 1 | cis | 29 | 0 |
| 1 | cis | 33 | 0 |
| ... | ... | ... | ... |

Long format is convenient for plotting and many longitudinal models.

## Outcomes and explanatory variables

- Outcome: plant height, measured in millimeters.
- Explanatory variables:
  - Strain ("cis" or "108")
  - Time: days since planting, or estimated age since emergence

## Assumptions

### Biological considerations

- True plant height is nonnegative.
- Height generally increases during the growth period.
- Growth eventually approaches a plateau.
- Emergence occurs between observation times.
- Individual plants may follow different trajectories.

### Longitudinal dependence

Repeated heights from the same plant are not generally independent. A model must account for relevant dependence.

Possible sources of variation:

- Differences between individual plants
- Temporal fluctuations within plants
- Shared environmental influences
- Measurement error

The appropriate covariance structure is not known in advance.

## Your task: preliminary analysis

- Exploratory data analysis
  - Identify plants, strains, and measurement times.
  - Check missing values and unusual observations.
  - Summarize heights by strain and time.
  - Examine emergence times and mature heights.
- Informative visualizations
  - Individual height trajectories by strain
  - Mean height trajectories by strain
  - Growth increments between measurements
  - Distribution of mature heights
  - Distribution of estimated emergence times

  - Consider both time scales: time since planting and time since emergence; (Which time scale seems most informative?)
- A proposed plan for formal analysis
  - A precise primary scientific question
  - An estimand for comparing strains
  - A suitable statistical method and corresponding assumptions required by that method (You are not required to reproduce the textbook's Gaussian-process model. The best analysis is not necessarily the most complicated one. Focus on reasoning, not model complexity.)

## Potental statistical models

Options may include:

- Gaussian-process models (used by the textbook)
- Mixed-effects models
- Generalized additive mixed models
- Functional data analysis
- Generalized estimating equations
- Neural networks (for a large number of plants)
- ...

These methods differ in assumptions, interpretation, and inferential goals.